

Problem Solving (MATH 1000)

Summer 2018

Thursday, 06/14/2018

Quiz 4: Section 10.4, 13.1, 13.2

Name (Print): _____

Score: _____ / 100

1. Oh, the difference a dollar makes! Suppose you run up a balance of \$10,000 on a credit card, with an interest rate of 22% compounded monthly.

- (a) 10 points If you only pay \$184.33 per month, how long will it take to pay off the balance?

$$\text{HINT: } T = \frac{\ln\left(\frac{M}{M - pP}\right)}{\ln(1 + p)}$$

To solve this question, we use the amortization formula:

$$M = P \frac{r(1 + r)^T}{(1 + r)^T - 1}.$$

Identifying that $M = \$184.33$, $P = \$10,000.00$, and (as 22% is an annual rate) $r = \frac{0.22}{12}$, we just solve for T as follows:

$$T = \frac{\ln\left(\frac{M}{M - pP}\right)}{\ln(1 + p)} = \frac{\ln\left(\frac{184.33}{184.33 - (0.22/12)(10,000)}\right)}{\ln(1 + 0.22/12)} \approx 287.33$$

So, since we get $T = 288$ when we round T up to the next whole month, it will take

$$288/12 = 24 \text{ years to pay off.}$$

- (b) 5 points How much interest do you pay over this time span?

To calculate the total interest paid, we simply take the monthly payments times the total number of months, then deduct the original balance. So, we get

$$(288 \text{ months})(\$184.33 \text{ per month}) - (\$10,000 \text{ original balance}) = \$43,087.04 \text{ interest paid.}$$

- (c) 10 points If you only pay \$183.34 per month, how long will it take to pay off the balance?

To solve this question, we use the amortization formula:

$$M = P \frac{r(1+r)^T}{(1+r)^T - 1}.$$

Identifying that $M = \$183.34$, $P = \$10,000.00$, and (as 22% is an annual rate) $r = \frac{0.22}{12}$, we just solve for T as follows:

$$T = \frac{\ln\left(\frac{M}{M - pP}\right)}{\ln(1 + p)} = \frac{\ln\left(\frac{183.34}{183.34 - (0.22/12)(10,000)}\right)}{\ln(1 + 0.22/12)} \approx 562.66$$

So, since we get $T = 563$ when we round T up to the next whole month, it will take

$$563/12 = 46.9166 \text{ years to pay off.}$$

Or, more precisely, 46 years, 11 months to pay off.

- (d) 5 points How much interest do you pay over this time span?

To calculate the total interest paid, we simply take the monthly payments times the total number of months, then deduct the original balance. So, we get

$$(563 \text{ months})(\$183.34 \text{ per month}) - (\$10,000 \text{ original balance}) = \$93,220.42 \text{ interest paid.}$$

- (e) 5 points If you only pay \$183.33 per month, how long will it take to pay off the balance? Will the balance ever be paid off?

HINT: If you end up trying to take the natural logarithm of a negative number when using the formula given in the first hint, then it means the balance will never be paid off.

To solve this question, we use the amortization formula: $M = P \frac{r(1+r)^T}{(1+r)^T - 1}$.

Identifying that $M = \$183.33$, $P = \$10,000.00$, and (as 22% is an annual rate) $r = \frac{0.22}{12}$, we can try to solve for T as follows:

$$T = \frac{\ln\left(\frac{M}{M - pP}\right)}{\ln(1 + p)} = \frac{\ln\left(\frac{183.33}{183.33 - (0.22/12)(10,000)}\right)}{\ln(1 + 0.22/12)} \approx \frac{\ln(-54,999)}{\ln(1.018333)}.$$

But, notice that this now tells us that we have to take the natural log of a negative number, which isn't possible! In fact, at this point, the payment is too small to keep up with the interest, and the balance will *never* be paid off!

2. Oh, the difference a percent makes! Suppose you run up a balance of \$10,000 on a credit card, with an interest rate of 21% compounded monthly.

- (a) 10 points If you only pay \$184.33 per month, how long will it take to pay off the balance?

To solve this question, we use the amortization formula: $M = P \frac{r(1+r)^T}{(1+r)^T - 1}$.

Identifying that $M = \$184.33$, $P = \$10,000.00$, and (as 21% is an annual rate) $r = \frac{0.21}{12}$, we just solve for T as follows:

$$T = \frac{\ln\left(\frac{M}{M - pP}\right)}{\ln(1 + p)} = \frac{\ln\left(\frac{184.33}{184.33 - (0.21/12)(10,000)}\right)}{\ln(1 + 0.21/12)} \approx 171.973$$

So, since we get $T = 172$ when we round T up to the next whole month, it will take

$$172/12 \approx 14.33 \text{ years to pay off.}$$

Or, more precisely, 14 years, 4 months to pay off.

- (b) 5 points How much interest do you pay over this time span?

To calculate the total interest paid, we simply take the monthly payments times the total number of months, then deduct the original balance. So, we get

$$(172 \text{ months})(\$184.33 \text{ per month}) - (\$10,000 \text{ original balance}) = \$21,704.76 \text{ interest paid.}$$

- (c) 10 points If you only pay \$183.34 per month, how long will it take to pay off the balance?

To solve this question, we use the amortization formula:

$$M = P \frac{r(1+r)^T}{(1+r)^T - 1}.$$

Identifying that $M = \$183.34$, $P = \$10,000.00$, and (as 21% is an annual rate) $r = \frac{0.21}{12}$, we just solve for T as follows:

$$T = \frac{\ln\left(\frac{M}{M - pP}\right)}{\ln(1 + p)} = \frac{\ln\left(\frac{183.34}{183.34 - (0.21/12)(10,000)}\right)}{\ln(1 + 0.21/12)} \approx 178.128$$

So, since we get $T = 179$ when we round T up to the next whole month, it will take

$$179/12 \approx 14.9167 \text{ years to pay off.}$$

Or, more precisely, 14 years, 11 months to pay off.

- (d) 5 points How much interest do you pay over this time span?

To calculate the total interest paid, we simply take the monthly payments times the total number of months, then deduct the original balance. So, we get

$$(179 \text{ months})(\$183.34 \text{ per month}) - (\$10,000 \text{ original balance}) = \$22,817.86 \text{ interest paid.}$$

- (e) 10 points If you only pay \$183.33 per month, how long will it take to pay off the balance?

To solve this question, we use the amortization formula:

$$M = P \frac{r(1+r)^T}{(1+r)^T - 1}.$$

Identifying that $M = \$183.33$, $P = \$10,000.00$, and (as 21% is an annual rate) $r = \frac{0.21}{12}$, we just solve for T as follows:

$$T = \frac{\ln\left(\frac{M}{M - pP}\right)}{\ln(1 + p)} = \frac{\ln\left(\frac{183.33}{183.33 - (0.21/12)(10,000)}\right)}{\ln(1 + 0.21/12)} \approx 178.194$$

So, since we get $T = 179$ when we round T up to the next whole month, it will take

$$179/12 \approx 14.9167 \text{ years to pay off.}$$

Or, more precisely, 14 years, 11 months to pay off.

3. 10 points Give the formula for the N th Fibonacci number in terms of previous Fibonacci numbers and write the first eight numbers in the Fibonacci sequence.

Give the formula for the N th Fibonacci number in terms of previous Fibonacci numbers is as follows, where F_k denotes the k^{th} Fibonacci number:

$$F_N = F_{N-1} + F_{N-2}.$$

The first eight numbers in the Fibonacci sequence are:

$$1, 1, 2, 3, 5, 8, 13, 21.$$

4. 5 points State the golden property and write the exact value of the golden ratio (not a decimal approximation of it).

The Golden Property is

$$\phi^2 = \phi + 1.$$

The exact value of the golden ratio is

$$\phi = \frac{1 + \sqrt{5}}{2}.$$

5. 10 points Using your answer from 4, write the exact value of ϕ^2 (not a decimal approximation of it).

The exact value of ϕ^2 can be found as follows:

$$\begin{aligned}\phi^2 &= \phi + 1 \\ &= \left(\frac{1 + \sqrt{5}}{2} \right) + 1 \\ &= \frac{1 + \sqrt{5}}{2} + \frac{2}{2} \\ &= \frac{3 + \sqrt{5}}{2}\end{aligned}$$